

Drone localization and trajectography through acoustic measurements

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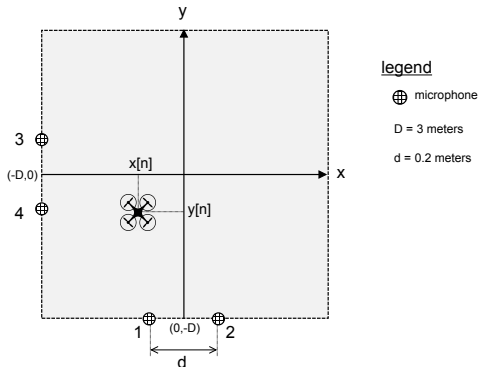
Problem statement

A drone moves in a plane for about 15 seconds, going back and forth between 2 positions about 1 meter apart.

At the same time, the sound it makes is recorded synchronously by 4 microphones at a sampling frequency of $f_s = 44.1$ kHz.

The aim is to estimate in real-time and as accurately as possible the trajectory of the drone by correctly processing the available acoustic signals.

Data



system 1 drone moving linearly in a plane

sensors 2 perpendicular antennas of 2 microphones

meaurements 4 acoustic signals synchronously recorded (.wav)

objective estimate the drone trajectory